Product Introduction

ZDH3070 is a single-chip, fully integrated RF front-end IC. It integrates a power amplifier (PA), a low-noise amplifier (LNA), and an RF switch (SW). It typically operates from a 3.3V or 5V supply, with a maximum saturated output power of 25dBm at the TX terminal and a typical noise figure of 1.6dB at the RX terminal. The input and output impedances are internally matched to 50Ω, resulting in a simple external application circuit.

ZDH3070 is packaged in a standard green lead-free QFN3x3-16 package, offering excellent reliability, cost-effectiveness, and a high performance-to-price ratio. It is suitable for improving the sensitivity and extending the transmission distance of UWB transceiver systems.

Typical application scenarios

- Precise real-time positioning systems
- Distance-aware wireless networks
- IEEE 802.15.4-2011 UWB transceivers

Absolute Maximum Ratings

Parameter	Value
Storage Temperature	-65°C~+150°C
Operating Temperature	-55°C~+125°C
Operating Voltage	6V
DC Control Input Voltage	5V
TX RF Input Power	+15dBm
ANT Input Power	+10dBm

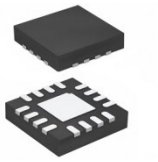
Operating status control logic table

State	TXEN	RXEN
TX on	H	L
RX on	L	H
Shutdown	L	L

"H": High level = +2.8V~VDD

"L": Low level = 0~+0.5V

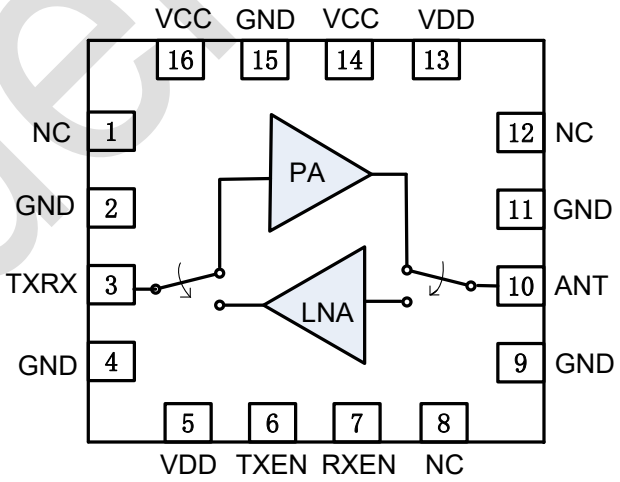
Features



- 3.3V or 5V single-voltage supply
- TX small-signal gain: 21dB @ 5500MHz
- TX saturated output power: 22dBm @ 5500MHz
- TX/RX port multiplexing, single antenna port
- Single-chip IC integrating power amplifier (PA), low-noise amplifier (LNA) and two single-pole double-throw switches
- Input/output 50Ω impedance matching
- Green lead-free 16-pin QFN3x3 package

This product complies with all relevant regulations and is halogen-free

Pin diagram (Top View)



PIN No.	Pin name	Description
1,8,12	NC	Empty, floating or grounded
2,4,9,11,15	GND	Ground
3	TXRX	TXRX Transceiver RF signal nput/output
6	PEN	Transmit enable
7	LEN	Receive enable
10	ANT	Antenna RF signal input/output
5,13	VDD	RX Power supply voltage input
14,16	VCC	TX Power supply voltage input
17	EPAD	Bottom GND

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Recommended operating conditions

Parameter	Unit	Minimum	Typical	Maximum
Operating voltage	V	2.8	3.3	5
Control voltage "1"	V	2.8	3.0	VDD
Control voltage "0"	V	0	-	0.5

TX Electrical Parameters

1. Test conditions: VDD=VCC=+3.3 V, Temp=+25° C, Freq: 4600MHz~6500MHz, PEN=3.0V, LEN=0V, 50Ω test system.

Parameter	Condition	Typical value					Unit
Frequency Range (F)		4600	5000	5500	6000	6500	MHz
Saturated output power (Psat)	Pin=5dBm	22	22	22	23	23	dBm
Gain	Pin=0dBm	21	20	21	22	19	dB
Input Return Loss (S11)	-	-5	-4	-3	-6	-9	dB
Output Return Loss (S22)	-	-4	-3	-4	-12	-12	dB
P1dB	-	22	21	22	23	23	dBm
TX Quiescent Current (Icq)	No RF input signal	40	40	40	40	40	mA
TX Operating Current (Icc)	-	199	224	237	206	177	mA
Efficiency	-	16	14	14	19	22	%

2. Test conditions: VDD=VCC=+5V, Temp=+25° C, Freq: 4600MHz~6500MHz, PEN=3.0V, LEN=0V, 50Ω test system.

Parameter	Condition	Typical value					Unit
Frequency Range (F)	-	4600	5000	5500	6000	6500	MHz
Saturated output power (Psat)	Pin=5dBm	23	22	22	24	25	dBm
Gain	Pin=0dBm	23	21	22	22	19	dB
Input Return Loss (S11)	-	-5	-3	-4	-7	-9	dB
Output Return Loss (S22)	-	-4	-3	-4	-15	-14	dB
P1dB	-	22	21	22	24	25	dBm
TX Quiescent Current (Icq)	No RF input signal	52	52	52	52	52	mA
TX Operating Current (Icc)	-	217	227	241	241	223	mA
Efficiency	-	17	12	13	22	29	%

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RX Electrical Parameters

1. Test conditions: VDD=VCC=+3.3 V, Temp=+25° C, Freq: 4600MHz~6500MHz, PEN=0V, LEN=3.0V, 50Ω test system.

Parameter	Typical value					Unit
Frequency Range (F)	4600	5000	5500	6000	6500	MHz
Gain	26	24	21	19	18	dB
Input Return Loss (S11)	-3	-2	-2	-2	-3	dB
Output Return Loss (S22)	-8	-4	-3	-3	-4	dB
RX 静态电流 (I_{DQ})	21	21	21	21	21	dB
RX O perating Current (I_{DD})	26	27	31	38	42	mA
P1dB	11	11	11	12	13	dBm
Noise Figure	1.6	1.8	2.0	2.0	2.0	dB

2. Test conditions: VDD=VCC=+5V, Temp=+25° C, Freq: 4600MHz~6500MHz, PEN=0V, LEN=3.0V, 50Ω test system.

Parameter	Typical value					Unit
Frequency Range (F)	4600	5000	5500	6000	6500	MHz
Gain	27	25	22	20	19	dB
Input Return Loss (S11)	-3	-2	-2	-2	-3	dB
Output Return Loss (S22)	-8	-4	-3	-3	-4	dB
RX Quiescent Current ($_{IDQ}$)	27	27	27	27	27	dB
RX O perating Current (I_{DD})	46	39	48	52	49	mA
P1dB	15	15	15	15	14	dBm
Noise Figure	1.6	1.8	2.0	2.0	2.0	dB

Chip shutdown electrical parameters

Test conditions: VDD=VCC=+3.3 V, Temp=+25°C, PEN=0V, LEN=0V, 50Ω test system.

Parameter	Condition	Minimum	Typical	Maximum	Unit
Leakage current	-	-		10	uA
TXRX-ANT isolation	-	-	-40	-	dB
ANT-TXRX isolation	-	-	-40	-	dB
TX-RX switching time	10%RF-90%RF/ 90%RF-10%RF	-	300	-	ns
Switching time after shutdown/on	50%Vctrl-10%RF/50%Vctrl-90%RF	-	300	-	ns

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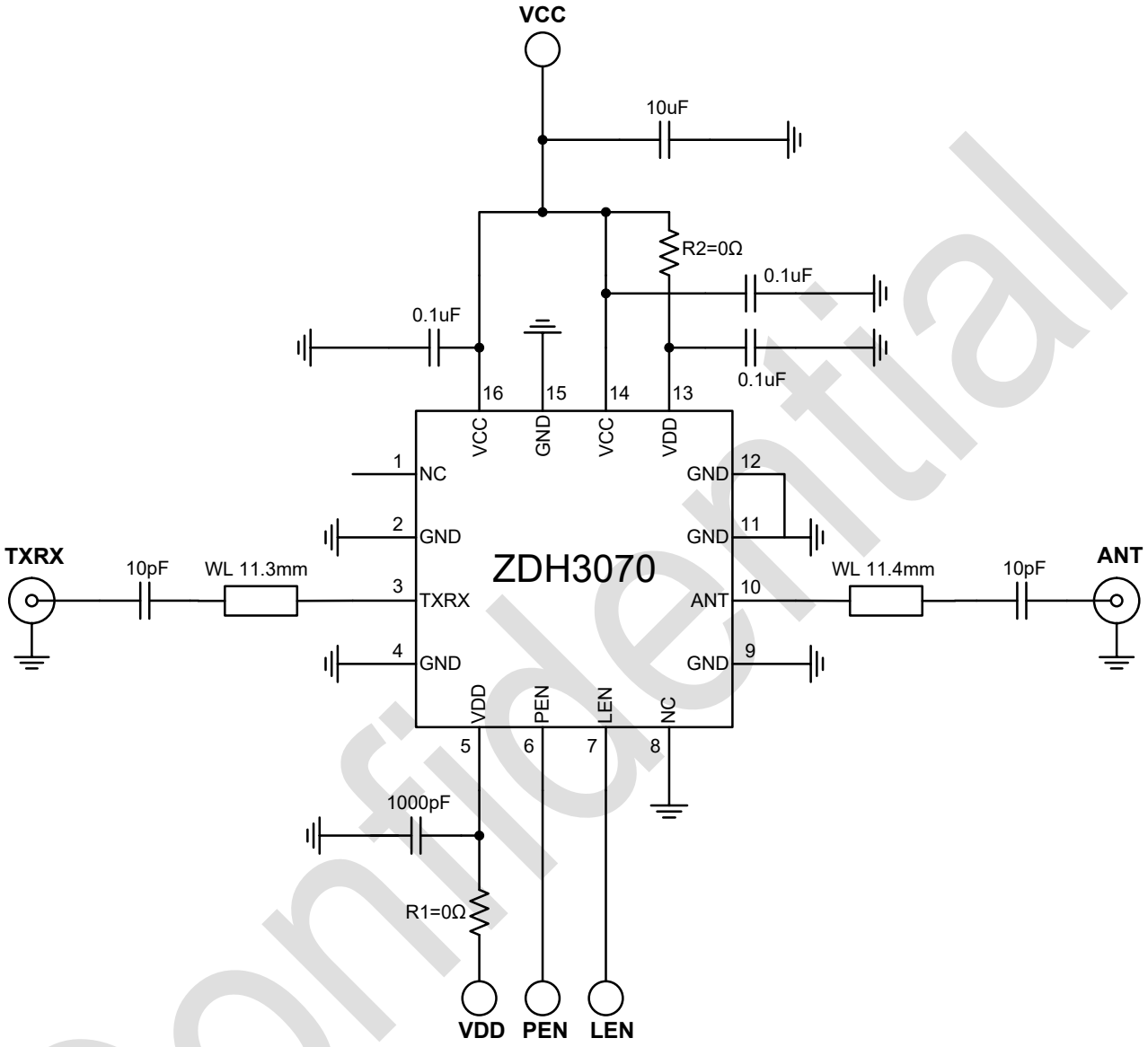
Application circuit diagram



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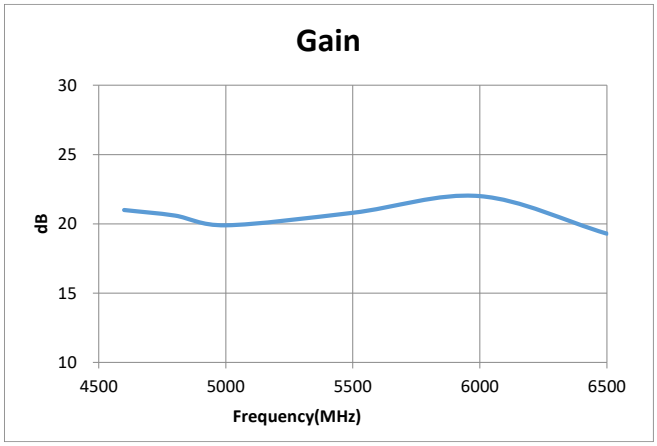
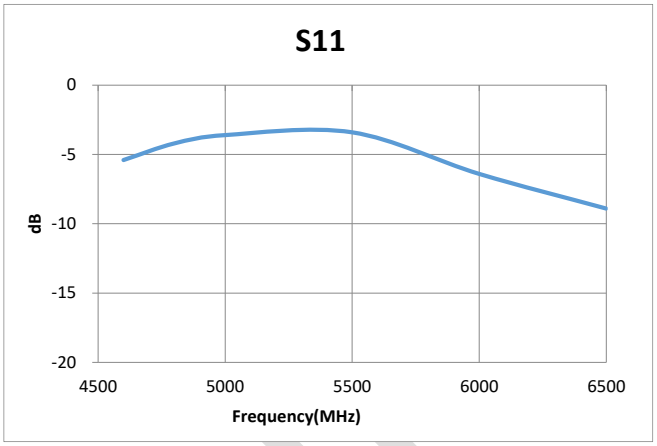
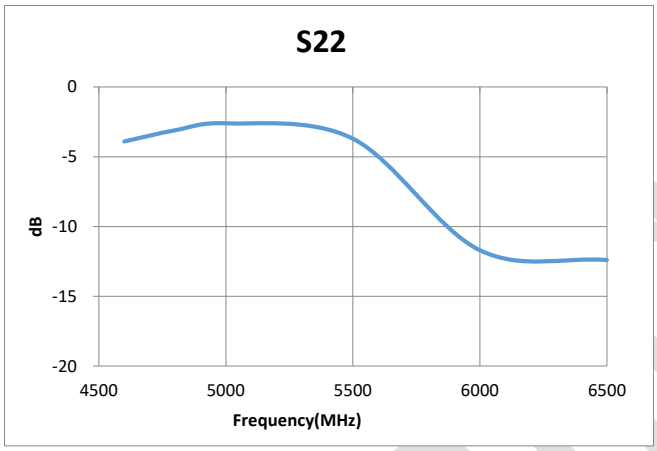
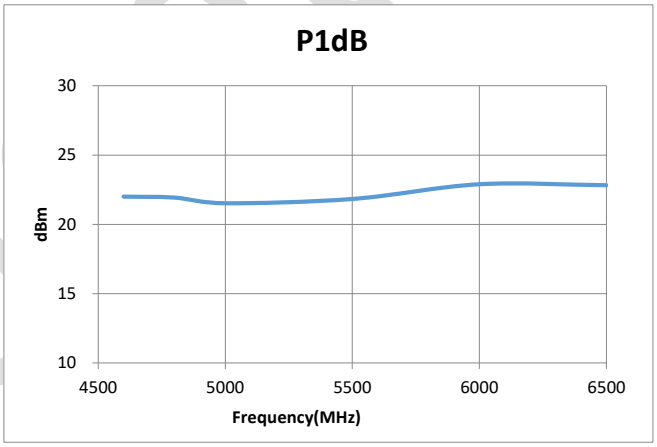
Typical performance curves (TX)

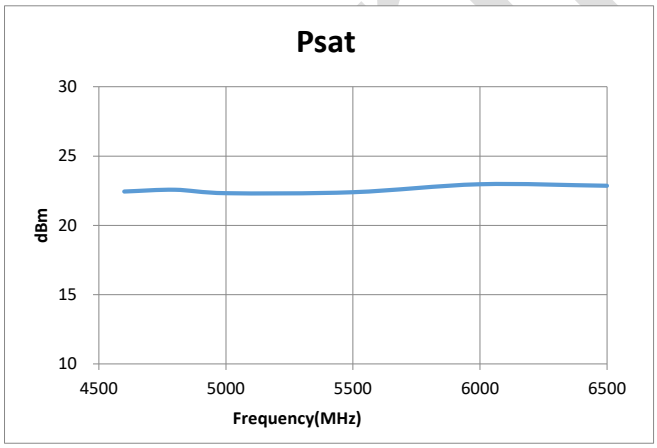
1. Test conditions: VDD=VCC=+3.3 V, Temp=+25° C, Freq: 4600MHz~6500MHz, PEN=3.0V, LEN=0V, 50Ω test system.

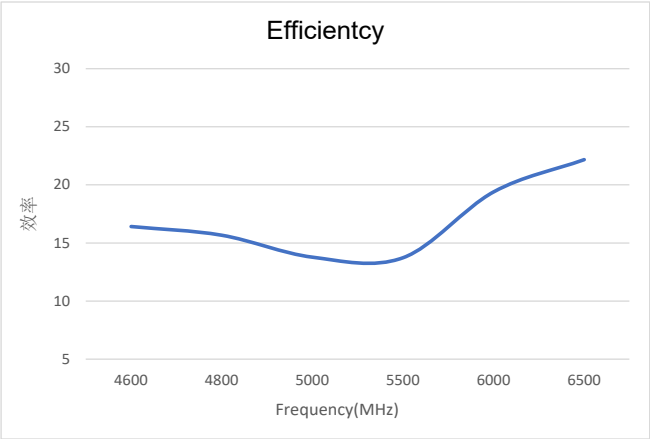






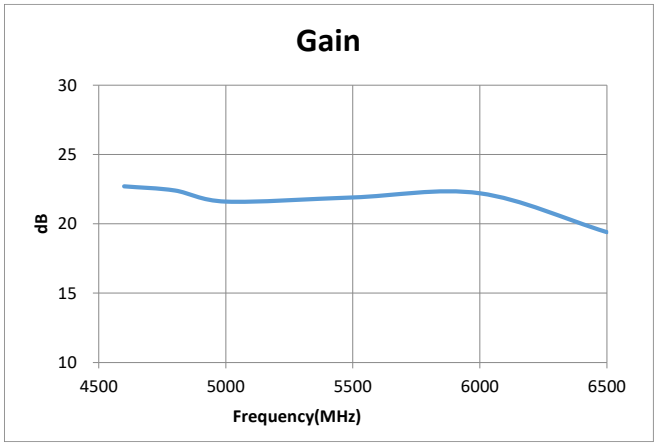


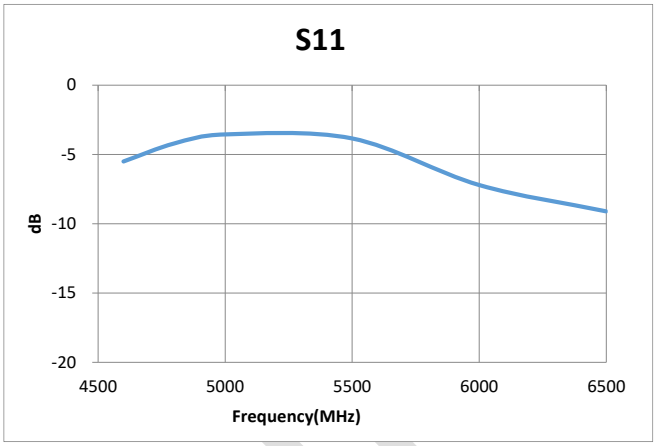


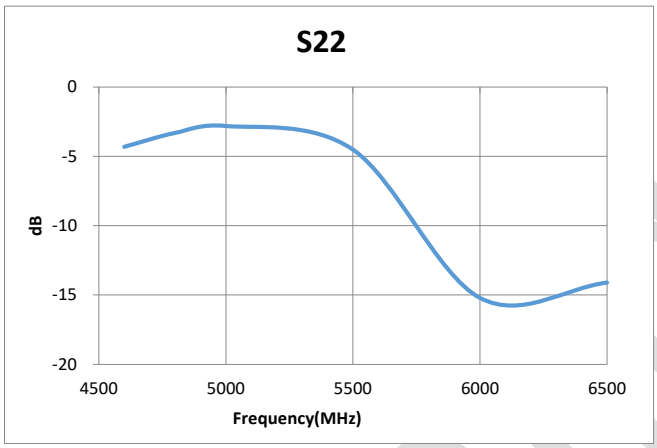


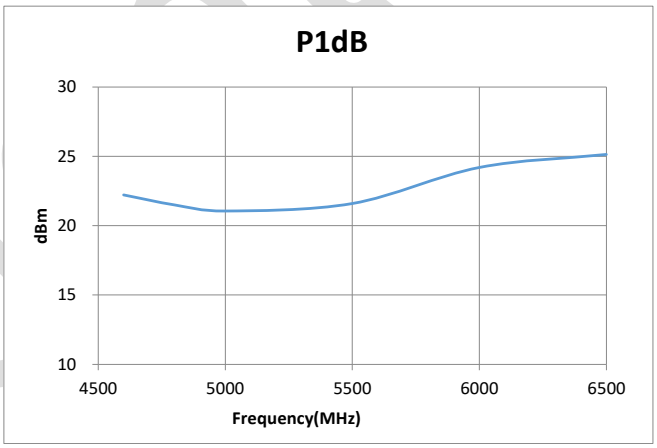
Typical performance curves (TX)

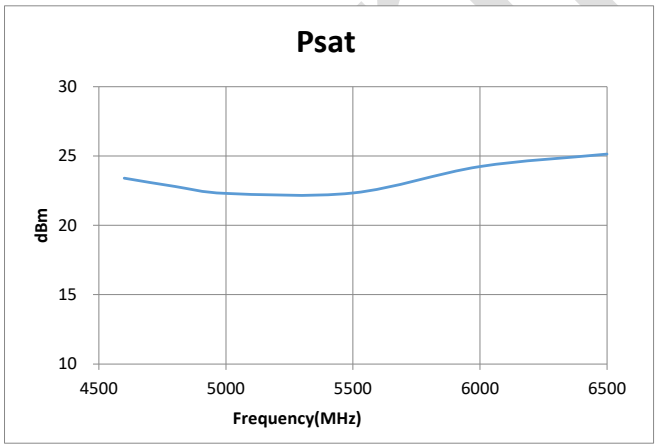
2. Test conditions: VDD=VCC=+5 V, Temp=+25° C, Freq: 4600MHz~6500MHz, PEN=3.0V, LEN=0V, 50Ω test system.

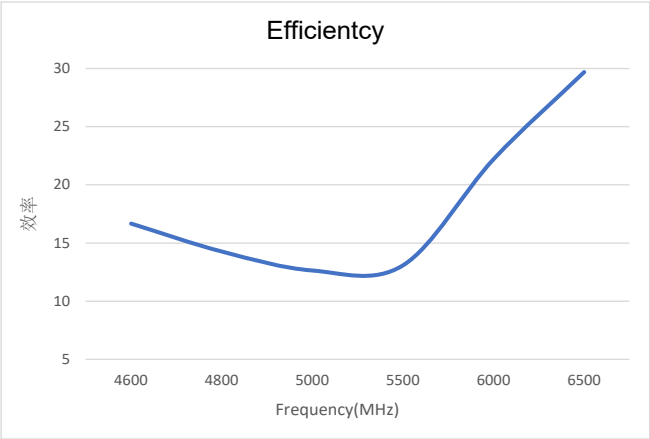






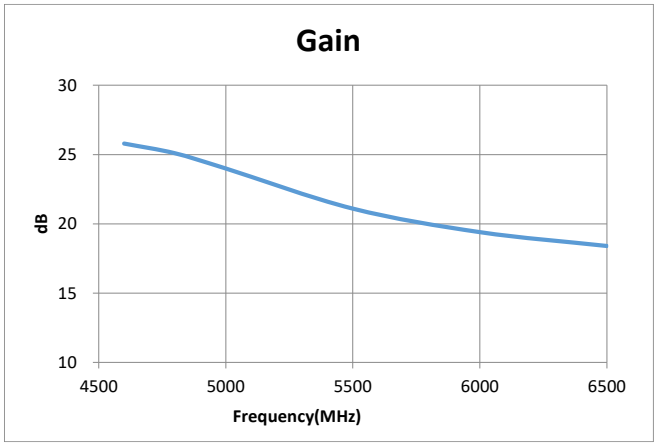


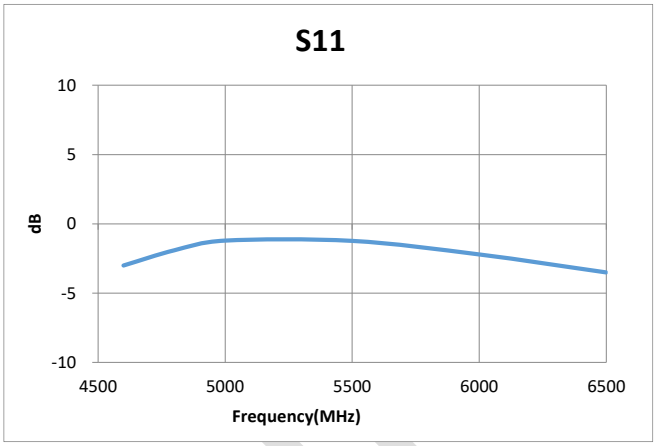


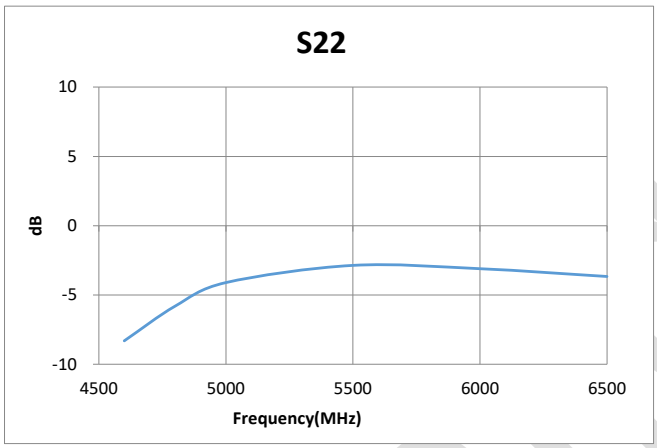


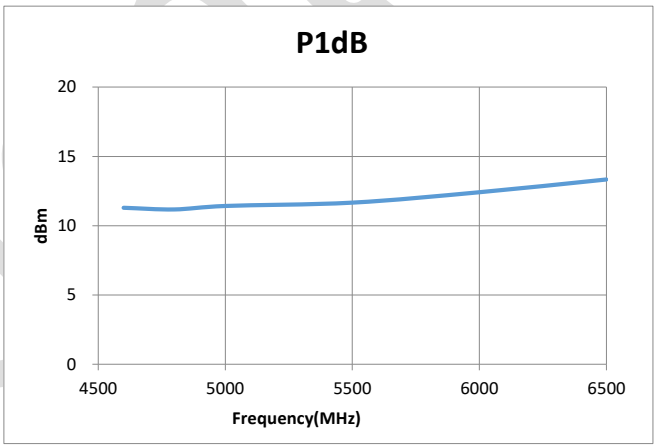
Typical performance curves (RX)

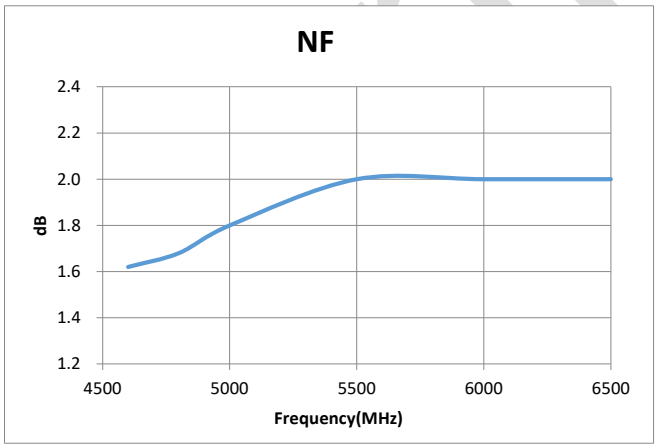
1. Test conditions: VDD=VCC=+3.3 V, Temp=+25°C, Freq: 4600MHz~6500MHz, PEN=0V, LEN=3.0V, 50Ω test system.





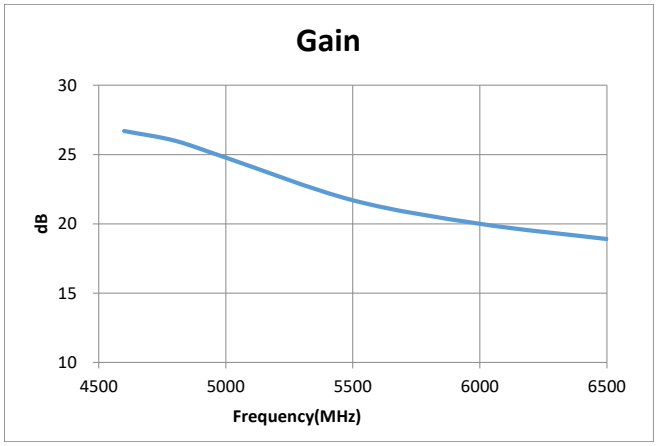


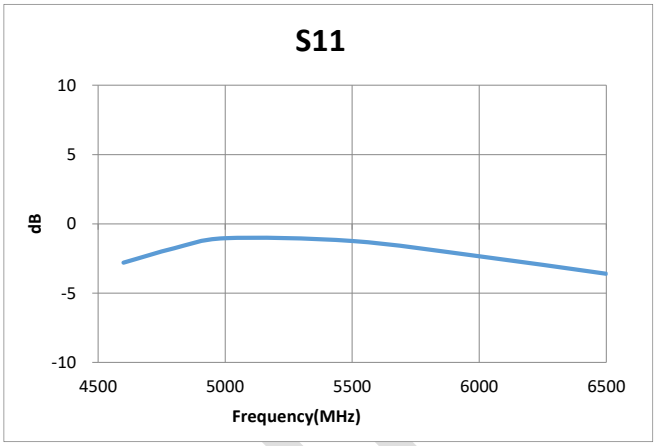


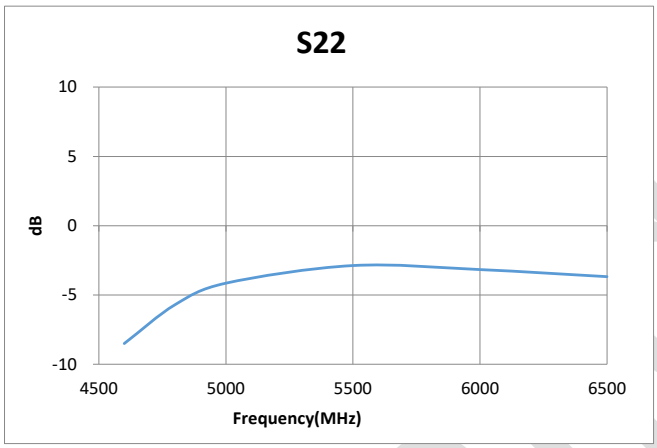


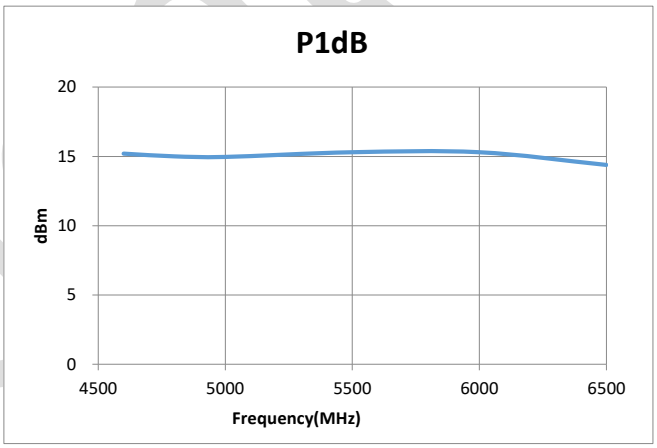
Typical performance curves (RX)

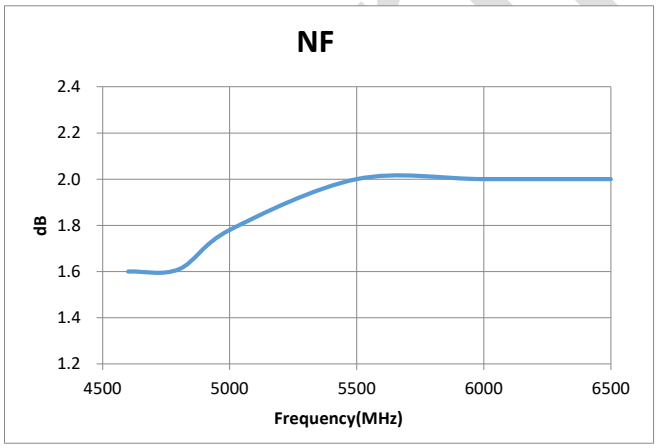
2. Test conditions: VDD=VCC=+5V, Temp=+25°C, Freq: 4600MHz~6500MHz, PEN=0V, LEN=3.0V, 50Ω test system.



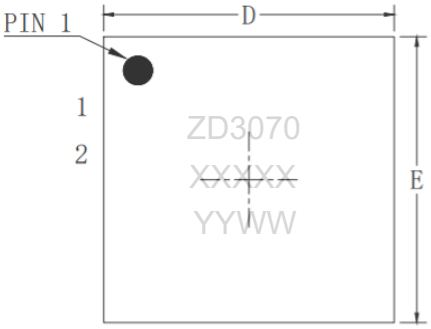




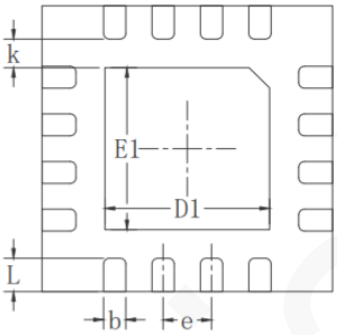




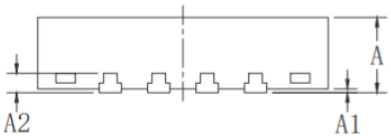
Package diagram



top view



bottom view



side view

symbol	Dimensions unit: mm		
	minimum value	Standard value	maximum value
A	0.700	0.750	0.800
A1	0.000	--	0.050
A2	0.195	0.203	0.211
D	2.950	3.000	3.050
E	2.950	3.000	3.050
D1	1.600	1.700	1.800
E1	1.600	1.700	1.800
k	0.300 Min.		
b	0.180	0.230	0.280
e	0.500 Typ.		
L	0.300	0.350	0.400

Ordering Information

Model	Marking	Packaging
ZDH3070	ZD3070	QFN3x3-16

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